KERAUNOS

Electromagnetic Launch from the Moon, Mars, and Earth

Maglev Space Launch System · SELENE · BRONTE · ASTERIA · ARES

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Perpetual Technologies · Engineering Whitepaper

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0. Executive Summary

Earth, with its thick atmosphere, is the hardest body in the system to launch from electromagnetically. So the program starts on the Moon, where the machine is native, and works outward. The full Earth injector is the long-horizon prize. It is not the opening move.

Launcher	Body	Phase	Character
SELENE	Moon	1, Proof of concept	No atmosphere: no tube, no plasma window, no aero-shell. Starter track 1–3 km; at 100 g, 2.9 km clears lunar escape in 2.4 s. The first segment is a working exporter, not a demo.
BRONTE	Earth	2–3, Chimborazo + ASTERIA, in parallel	A 14 km first build on a ~21.3 km straight through-mountain line at Chimborazo, Ecuador; tunnel muzzle at 5,900 m. Proves tube, plasma window, and aero-shell at sub-scale. DoD hypersonic test contracts pay for it.
ASTERIA	Airless bodies	2–3, (concurrent with BRONTE)	SELENE-class launchers stamped onto Phobos, Deimos, Ceres, the Belt. The core bet, runs alongside Chimborazo, not after it.
ARES	Mars	4, The port	Mars atmosphere is 0.6% of Earth's; the tube is featherweight. ARES closes the return-trip problem every settlement plan dies on.
BRONTE full	Earth	5, The endgame	The ~21.3 km straight Mach-11 line, then beyond, built only after Chimborazo Phase 1 validates cost and subsystems.

SELENE: Moon (Phase 1)

Parameter	Value	Notes
Track (PoC)	1–3 km, 50–100 g	At 100 g: 1.5 km = orbit speed (1.72 km/s); 2.9 km = escape (2.38 km/s) in 2.4 s
Track (cargo)	10–12 km, 30 g	2.43–2.66 km/s; escape with margin; ~9 s ride
Track (passengers)	~50 km, 3 g	1.72 km/s to low lunar orbit in 58 s (escape-rated 3 g line: ~96 km, future)
Energy per kg	0.41 kWh orbit / 0.78 kWh escape	Orbit ≈ 1/22 of Earth's 8.9 kWh/kg-to-LEO; escape ≈ 1/11
Atmosphere	None	No tube, no plasma window, no aero-shell. The track is the launcher.
Cooling	20 K closed-loop cryoplant	Sealed cryostats hold every coil at 20 K; shadowed cold sinks and ~250 K regolith at 2 m depth help, sunlit surfaces swing far hotter, so the cryostat does the real work

BRONTE + ASTERIA: Earth & Airless Bodies (Phase 2–3, Chimborazo Project, concurrent with vacuum-world deployment)

Parameter	Value	Notes
Site	Chimborazo, Ecuador	Summit 6,263 m; 1.47°S; massif sits on a 4,000 m plateau
Phase-1 build	14 km, portal ~2,920 m	Eastern (upper) 14 km of the full straight line; lower-flank portal
Exit	Tunnel muzzle 5,900 m, 12.3° east	Single straight 12.3° grade, no curve at all; ~363 m below peak elevation
Exit air	ρ ≈ 54% SL (p ≈ 47%)	Peak drag and heating roughly halved; weaker ground-level boom
Equatorial bonus	+465 m/s free	Fires east; net launcher requirement to LEO ~7.5 km/s, not ~8
Performance	Mach 9.1 (30 g) / 2.9 (3 g)	Full ~21.3 km straight line later: Mach 11.2 cargo / 3.5 at human-rated 3 g
Revenue	$50K–$500K per test run	Hypersonic test services from day one, the market Auriga Space is already selling into

ARES: Mars (Phase 4)

Parameter	Value	Notes
Atmosphere	~6 mbar (0.6% of Earth)	Featherweight tube; the planet does 99.4% of the plasma window's job
Track (orbit)	~180 km, 3.3 g	Low Mars orbit (~3.4 km/s) in a 105 s run
Track (escape)	~390 km, 3.3 g	Mars escape (~5.0 km/s) in a 155 s run
Energy per kg	1.6 kWh orbit / 3.5 kWh escape	Between Moon and Earth in scale; far easier than Earth in engineering
Staging	Phobos and Deimos	Catch points and depots; SEP tugs fly the interplanetary legs

1. SELENE: The Moon Machine

The Moon has no atmosphere. That sentence deletes every hard engineering problem on the Earth list.

On the Moon these things do not exist: the vacuum tube (the track is already in vacuum), the plasma window (no atmosphere to keep out), the aero-shell, the blast shutters, the transpiration cooling. The track lays on regolith, braced and aligned. That is the system.

The starter track is **1–3 km**, airport-runway length. At 100 g, a 1.5 km run reaches lunar orbit speed (1.72 km/s); a 2.9 km run clears escape (2.38 km/s) in 2.4 seconds. Water, regolith, and metal don't mind the g-load. The very first segment isn't a demo. It is a working exporter.

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SELENE track from a = v²/2L ⇒ v = √(2aL), launch a = 100 g = 981 m/s²:
Orbit: L = 1.5 km ⇒ v = √(2 × 981 × 1,500) = 1,716 m/s = 1.72 km/s (low lunar orbit)
Escape: L = 2.9 km ⇒ v = √(2 × 981 × 2,900) = 2,385 m/s = 2.38 km/s, t = v/a = 2.4 s
Launch energy: E = ½v² = ½ × 2,385² = 2.84 × 10⁶ J/kg = 0.79 kWh/kg (vacuum: no tube, no kick to escape)
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0.78 kWh per kg to escape. Lunar escape costs roughly one-eleventh of Earth-to-LEO launch energy (0.78 vs 8.9 kWh/kg at 8 km/s); lunar orbit, about one-twenty-second. From the Moon you are already partway to everywhere: a Mars-transfer trajectory from the lunar surface costs ~3.0 km/s total, about **1.25 kWh/kg**, roughly one-seventh of Earth-to-LEO energy, and roughly one-fifteenth the energy of flying the same trip up from Earth's surface.

The south pole is the right site. Shackleton-rim and Connecting-Ridge sites see sun 80–90% of the time; permanently shadowed cold traps sit next door; Earth hangs permanently on the horizon for communications. Rev 2 honesty on two Rev 1 claims: polar *surface* temperatures swing widely (shadow cryo-cold to ~200 K+ in sun), so the 20 K coils live in sealed cryostats regardless, the real thermal gift is vacuum, cold sky, and ~250 K regolith two meters down. And the ice: the famous "600 million tonnes" belongs to *north-pole* crater radar (Mini-SAR 2010). For Shackleton itself the honest statement is bounded, radar and reflectance permit up to ~5–10 wt% ice in wall regolith, with no credible tonnage yet; Chang'e-7 (launch ~Aug 2026) goes to ground-truth exactly this. The architecture survives the uncertainty: regolith, oxygen, and metal exports close the early business even in the lean-ice case.

The physics has been settled for fifty years. Gerard O'Neill's NASA Ames summer studies (1976–77) developed the lunar mass driver; Mass Driver One was demonstrated in 1977. A San José State University engineering study (E. Miller, 2023) sizes a polar machine launching 25.4 kg pellets at 2.4 km/s every 10 seconds on 8.7 MW of solar power, ~1.0 MA per bucket coil, SELENE's starter track is exactly that class of machine. What killed every previous program was the same thing: all government funding, no revenue until a Moon base already existed. Cislunar freight removes that deadlock.

The product is propellant, and mass itself. Polar ice (wherever the surveys land), electrolyzed to LOX/LH2 or processed with imported carbon to methalox, becomes propellant for every vehicle in cislunar space, delivered to L1/L2 depots at roughly 10× less than lifting the same kilogram from Earth. Regolith is radiation shielding; sintered metal is structure. The Moon doesn't need factories to matter, it needs to be a source of cheap mass, and it already is.

SELENE grows by ladder, 1–3 km PoC → 10–12 km / 30 g hardened cargo → ~50 km / 3 g passenger line, each step funded by the previous segment's exports. The SELENE kit is the template for every airless body worth working: Phobos, Deimos, Ceres, the Belt. Design once, stamp copies. ASTERIA is the real bet. *(The SELENE South Pole Design Map, Annex A, follows this section.)*

The dust problem, and why a contactless launcher is the right answer

Lunar regolith is the nastiest material in the inner solar system to run machinery in: jagged, never weathered, electrostatically charged, and it gets into everything. It is a first-order design driver, not a footnote, and the south-pole site makes it worse.

> `Apollo ground truth: dust on radiators overheated Apollo 16 instruments (it kills emissivity); Apollo 12 suit seals leaked 0.25 psi/min vs a 0.30 limit; gauge faces abraded unreadable. NASA's 2005 tally: vision loss · false readings · clogged mechanisms · abrasion · thermal-control loss · seal failure · inhalation.`
> `Polar twist: near the terminator, UV / solar-wind charging drives ~`**`300 V`**` sunlit-vs-shadow potentials and `**`electrostatic dust levitation`**`; shadowed ground has `*`no`*` charge-dissipation path, dust clings and will not let go. SELENE's ridge sits in exactly this terminator zone.`

SELENE is unusually well-suited to fight it, because the launcher is contactless. Most Apollo dust failures were mechanical, seals, bearings, sliding contact. SELENE's drive has none in the load path:

- **Maglev = no rolling or sliding contact.** The pod rides a 200 mm vacuum gap; there are no wheels, rails, or bearings for dust to abrade or clog, the exact failure mode that wrecked Apollo hardware is simply absent on a contactless track.
- **Magnets sealed at 20 K.** The REBCO coils live in sealed cryostats; dust never reaches the conductors or the cold mass.
- **Electrodynamic Dust Shield (EDS) on the vulnerable surfaces.** Radiators and solar arrays are the real victims (dust drops emissivity and power). NASA's EDS, travelling-wave electric fields, no moving parts, cleared dust from glass and thermal radiators on Firefly's Blue Ghost lander in March 2025; SELENE coats its radiators, arrays, and optics with it, mounted vertically and elevated on the embankment.
- **Launch dust dome.** The EOS-1 pod's front heat shield doubles as a dust dome, containing the regolith lofted at release.
- **Charge management.** Grounded, conductive surfaces plus EDS bleed off the terminator's static, so dust has a path to leave rather than cling.

Built by robots, not boots. Dust is also a health hazard (Apollo "lunar hay fever"; the inhalation risk in the 2005 list), so SELENE is assembled mostly by machines: Optimus-class humanoid robots (Tesla's Optimus, NASA's dust-rated Valkyrie, Apptronik's Apollo) plus excavator and regolith-moving rovers, autonomous and teleoperated, with humans supervising and repairing rather than shovelling regolith by hand. The same EDS and sealed-joint approach protects the robots' own actuators. **Honest residuals:** radiators, optics, and robot joints still need active dust management for years, and the polar charged-dust environment is not yet flight-characterised, Chang'e and Artemis surface data will tighten it. (NASA EDS on Blue Ghost, Mar 2025; Apollo dust effects, NASA 2005; terminator charging ~300 V, lunar-surface-charging literature.)

What SELENE earns

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CAPEX (PoC): no tube, no plasma window, a short surface track, first coils, a ~10 MW solar plant,
cryo and radiators ride 1–2 heavy-lift flights → ~$1–3B, an order of magnitude below BRONTE's $10–
20B.
REVENUE (cislunar, sold in space, see §13):
Propellant / oxygen / structural metal to L1–L2 depots: demand-limited by the cislunar market
(~$1.8B/yr today, growing with depots, SEP tugs, and Mars staging) → ~$0.5–2B/yr near-term, scaling
toward multiple $B/yr.
He-3 (quantum-cryogenics luxury, §13.4): Interlune-scale ~1.3 kg/yr (10,000 L) × ~$20M/kg →
~$25M/yr, a high-margin kicker.
Launch electricity is ~$0.02–0.05/kg (0.41–0.78 kWh/kg), negligible; margin is set by capex
amortization and ISRU processing energy, not the launcher.
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SELENE is the better business: roughly **10× lower capex than BRONTE and revenue from day one** into a cislunar market that only grows, which is exactly why SELENE (Phase 1), not Earth, opens the program. (Cislunar market and He-3 figures from §13 deep-research pass; energy from §2.)